

COMMUNICATION SYSTEMS

Last-Minute Exam Pack • Derivations • Formula Sheet • Numericals • 2-Hour Revision

1. Most Important Derivations

A. AM Wave Equation ■■■

Carrier: $c(t) = A_c \cos(\omega_c t)$

Message: $m(t) = A_m \cos(\omega_m t)$

AM wave: $s(t) = A_c [1 + m \cos(\omega_m t)] \cos(\omega_c t)$

$m = A_m / A_c$

Expanded: $s(t) = A_c \cos \omega_c t + (mA_c/2) \cos(\omega_c + \omega_m)t + (mA_c/2) \cos(\omega_c - \omega_m)t$

Contains: Carrier + USB + LSB; $BW = 2f_m$

B. AM Power Derivation ■■■

$P_c = V_c^2 / (2R)$

$P_{USB} = P_{LSB} = (m^2/4)P_c$

$P_t = P_c (1 + m^2/2)$

$\eta = m^2 / (m^2 + 2)$; maximum AM efficiency = 33.3%

C. Carson's Rule ■■■

$BW = 2(\Delta f + f_m)$ and $\beta = \Delta f / f_m$

D. Sampling Theorem ■■■

Sampling frequency must satisfy: $f_s \geq 2f_m$. Below this limit, aliasing occurs.

E. Noise Power ■■■

$N = kTB$ where $k = 1.38 \times 10^{-23}$ J/K. Noise voltage: $V_n = \sqrt{4kTBR}$.

2. Important Formula Cheat Sheet

Topic	Formula
AM modulation index	$m = A_m / A_c = (V_{max} - V_{min}) / (V_{max} + V_{min})$
AM bandwidth	$BW = 2f_m$
AM total power	$P_t = P_c (1 + m^2/2)$
AM sideband power	$P_{USB} = P_{LSB} = (m^2/4)P_c$
FM modulation index	$\beta = \Delta f / f_m$
FM Carson bandwidth	$BW = 2(\Delta f + f_m)$
FM limits	$f_{max} = f_c + \Delta f$; $f_{min} = f_c - \Delta f$
Noise power	$N = kTB$
Noise factor	$F = (S/N)_{in} / (S/N)_{out}$
Noise figure	$NF(dB) = 10 \log_{10} F$
Sampling / Nyquist rate	$f_s \geq 2f_m$
PCM bit rate	$R_b = n f_s$
Image frequency (high-side injection)	$f_{image} = f_c + 2f_m$

3. Expected Numericals — With Solutions

1. FM Wave ■■■

Given: $\beta = 7$ and $\omega_m = 6 \times 10^4 \text{ rad/s}$.

$$f_m = \omega_m / (2\pi) = 9.55 \text{ kHz.}$$

$$\Delta f = \beta f_m = 7 \times 9.55 = 66.85 \text{ kHz.}$$

$$\text{Carson: BW} = 2(66.85 + 9.55) = \mathbf{152.8 \text{ kHz.}}$$

2. Thermal Noise ■■■

Given: $R = 600 \Omega$, $B = 20 \text{ kHz}$, $T = 300 \text{ K}$.

$$N = kTB = (1.38 \times 10^{-23})(300)(20000) = \mathbf{8.28 \times 10^{-14} \text{ W.}}$$

3. AM Modulation Index ■■■

Given: $V_{\max} = 8 \text{ V}$, $V_{\min} = 2 \text{ V}$.

$$m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min}) = 6/10 = \mathbf{0.6.}$$

4. Image Frequency ■■■

Given: $f_s = 600 \text{ kHz}$, $f_{IF} = 455 \text{ kHz}$.

$$f_i = f_s + 2f_{IF} = 600 + 910 = \mathbf{1510 \text{ kHz.}}$$

4. Last-Minute 2-Hour Revision Plan

0–20 min	AM	AM equation • power • modulation index • sidebands • high/low-level modulation
20–40 min	FM	FM equation • Carson rule • Armstrong method • Foster–Seeley discriminator • FM vs PM
40–60 min	Receivers	Superheterodyne • AGC • delayed AGC • image frequency • IFRR
60–80 min	Pulse Modulation	PAM/PWM/PPM • PCM block diagram • Delta Modulation • ADM • granular noise • slope overload
80–100 min	Sampling + Noise	Sampling theorem • aliasing • noise figure • noise factor • SNR
100–120 min	Numericals	FM • noise • AM • image frequency • sampling frequency

5. High-Priority Theory & Comparisons

Top theory: Superheterodyne receiver; Foster–Seeley discriminator; Armstrong FM generation; SSB generation; Delta Modulation + ADM.

Top numerical types: FM modulation index/Carson rule; thermal noise; AM power/modulation index; image frequency; sampling frequency.

AM vs FM: AM varies amplitude and is more noise-sensitive; FM varies frequency and has better noise immunity but generally needs more bandwidth.

PAM vs PWM vs PPM: PAM → amplitude varies; PWM → pulse width varies; PPM → pulse position varies.

FM vs PM: FM varies instantaneous frequency; PM varies instantaneous phase.

6. Exam Writing Checklist

- 10-mark theory: Definition → neat block diagram → working → equations → advantages/applications.
- Numericals: Given → Formula → Substitution → Final answer with units.
- Draw diagrams large and label every block clearly.
- Revise formulas once immediately before entering the exam.